Student Seminar - IDP Presentation 2

Can Ersoy - 06.10.2026

Topics from Previous Meeting

1) Discarded Golden Dataset Paper

2) Changing Search Query

Changing Search Query

New NCBI Query: “**peptide self-assembly**”

Includes paper: Self-Assembly of Block Heterochiral Peptides into Helical Tapes [1]

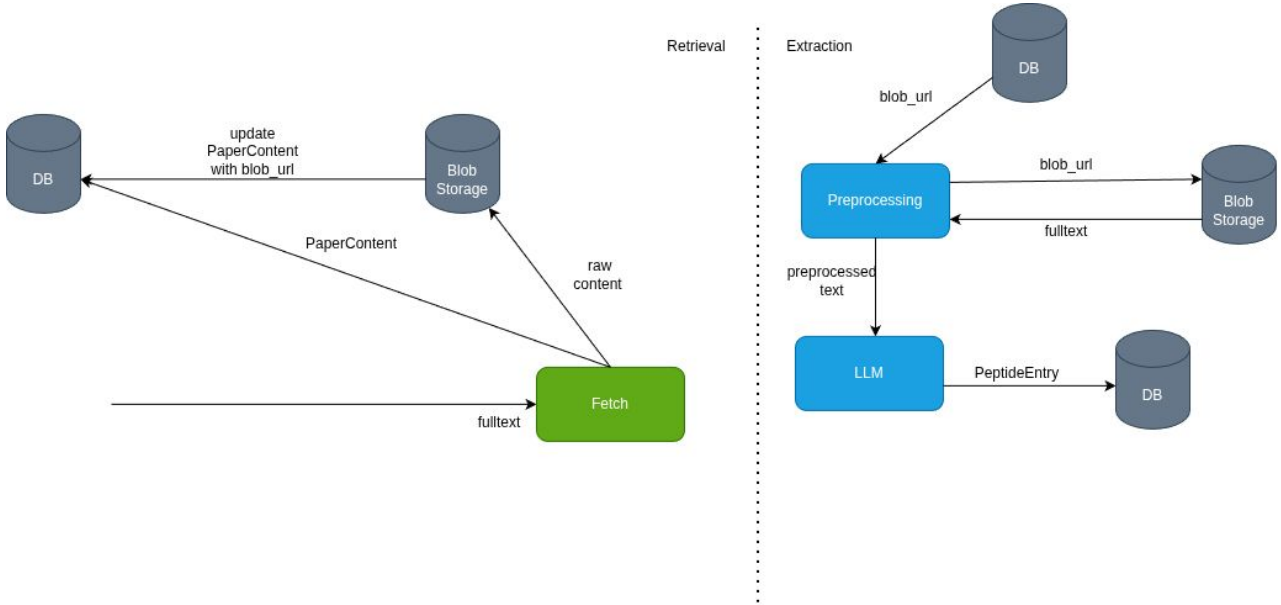
9631 results as of 30.09

[1]: Clover TM, O'Neill CL, Appavu R, Lokhande G, Gaharwar AK, Posey AE, White MA, Rudra JS. Self-Assembly of Block Heterochiral Peptides into Helical Tapes. J Am Chem Soc. 2020 Nov 25;142(47):19809-19813. doi: 10.1021/jacs.9b09755. Epub 2020 Apr 27. PMID: 32338879; PMCID: PMC7606833.

New Way to Store Papers

Full Text retrieval doesn’t scale well

Solution: Azure Blob Storage



Pdf Paper Full Text

OpenAlex can be used to find open-access pdf urls.

Tested 50 dois, only got 3 pdfs.

Some pdf urls are behind bot check (not programmatically accessible)

Extraction

1. Way: Pass full-text directly to LLM

2. Way: Pass only specific sections (Methods, Experiments, Materials etc.) -> similar to PeptideMiner

3. Way: Pass only paragraphs that include specific keywords

SYSTEM_PROMPT: You will receive the full text of a scientific paper and your goal is to extract experimental peptide self-assembly information according to the given schema. There can be several entries in a single paper be careful! If a field is not provided give "NP" for that field and write nothing else. Don't leave any fields empty

PEPTIDE_MINER_SYSTEM_PROMPT: polypeptide self-assembly literature information extraction

Example Paper PMID: 18070345

PeptideMiner (4 entries - 1 hydrogel 3 none)

sequence	length	type	n_terminal	c_terminal	temperature	ph
FF	2	none or hydrogel	Fmoc	Free	37 °C	7

Extraction (1 entry)

sequence	length	type	n_terminal	c_terminal	temperature	ph
FF	2	hydrogel	Fmoc	free	20	NP

PeptideMiner

peptide_concentration	solution	solvent	volume_ratio	incubation_period	peptide_conjugate_mixture
5.025 or 10.05 mg/ml	DMSO/PBS	Dispersion medium= PBS 1x, DMSO (gelling agent)= 100mg/ml	5/95 or similar	NP	Peptide

Extraction

peptide_concentration	solution	solvent	volume_ratio	incubation_period	peptide_conjugate_mixture
1.0 wt%	purified MilliQ water	DMSO	NP	< 10	peptide

PeptideMiner

conjugate_partner	linear_cyclic	detection_method	preparation_protocol
NP	Linear	Confocal Microscopy	Lyophilized Fmoc-Phe-Phe dipeptide was weighed and dissolved in DMSO. This was applied to the desired dispersion medium.

Extraction

conjugate_partner	linear_cyclic	detection_method	preparation_protocol
NP	linear	inversion test and confocal microscopy	Lyophilized Fmoc-Phe-Phe dipeptide was dissolved in dimethyl sulfoxide (DMSO) to create a gelling agent of 25 or 100 mg/ml, which was then applied to the dispersion medium to a final concentration of 0.5 to 1.0 wt%.

evidence_span
Hydrogel samples were created from a stock gelling agent of either 25 or 100 mg/ml. Successful gelling was performed in purified MilliQ water with both stock solutions. Final Fmoc peptide concentrations ranged from 0.5 to 1.0 wt% with successful hydrogel results.

Extraction Notes

- Pydantic field descriptions matter
- System prompt matters
- Assumptions matter
- Fine-tuning (?)

Next Steps

- Try different approaches for extraction and document the outcomes
- Get access to local LLM